

Unit 2.7 The derivative function

Note Title

2015/03/18

Definition of the derivative function:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (x)$$

$$f(x) = x^2 \longrightarrow$$

$$f'(x) = 2x$$

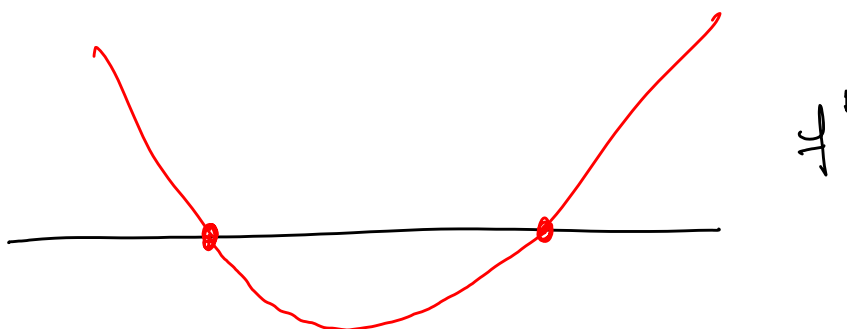
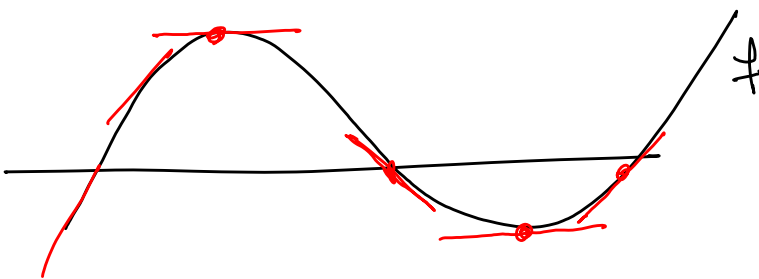
$$f(x) = \sin x \longrightarrow$$

$$f'(x) = \cos x$$

$$f(x) = e^x \longrightarrow$$

$$f'(x) = e^x$$

Graphically:



Continuity vs differentiability

f is continuous in $x=a$ if $\lim_{x \rightarrow a} f(x) = f(a)$

— you can sketch the function without lifting your pen

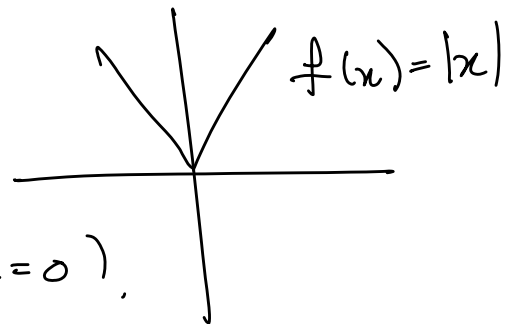
f is differentiable in $x=a$ if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists (is a finite number)

— you can draw a non-vertical tangent at $x=a$
tangent with a finite gradient
— the curve is smooth

Three ways of failing to be differentiable

• The curve makes a "sharp connection" at a point

Eg $f(x) = |x|$ is not differentiable in $x=0$.

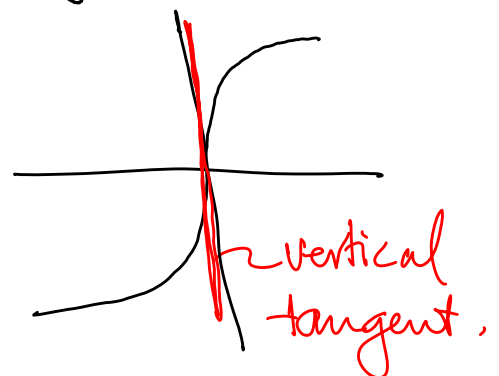


(although f is continuous in $x=0$).

• The curve has a vertical tangent at a point

Eg $f(x) = \sqrt[3]{x}$

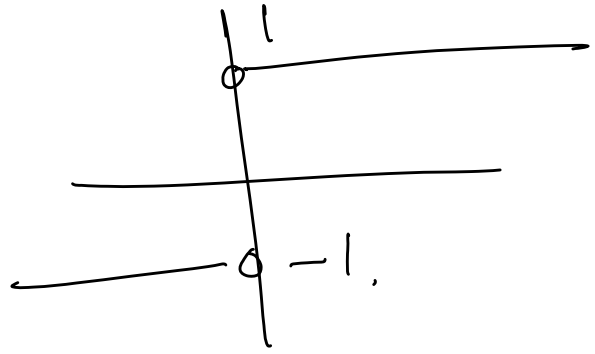
f is not differentiable in $x=0$.



- The function is discontinuous at a point

$$f(x) = \frac{\ln|x|}{x}$$

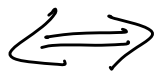
f is not differentiable
in $x=0$.



This holds if f has a jump, removable or infinite discontinuity

Logical & connected statements

You are over 16



You qualify for a SA ID

$$A \Leftrightarrow B$$

It rains



It is wet

$$A \Rightarrow B$$

$A \Rightarrow B$ does not mean $B \Rightarrow A$

It does mean $B^c \Rightarrow A^c$

NB Relationship between continuity and differentiability.

f is continuous in $x=a$



f is differentiable in $x=a$.

Diff $\Rightarrow$ Cont

$$(A \Rightarrow B)$$

Not Cont $\Rightarrow$ Not diff.

$$(B^c \Rightarrow A^c)$$

Theorem. If a function f is differentiable in $x=a$ then f is continuous in $x=a$.

Need to prove: $\lim_{x \rightarrow a} f(x) = f(a)$

What we know: $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists (is finite)

Proof: $\lim_{x \rightarrow a} f(x)$

$$= \lim_{x \rightarrow a} (f(x) - f(a) + f(a))$$

$$= \lim_{x \rightarrow a} \underbrace{(f(x) - f(a))}_{x-a} \cdot (x-a) + \underbrace{\lim_{x \rightarrow a} f(a)}_{f(a)}$$

AB/Pr.

$$= \underbrace{f'(a)}_k \cdot \underbrace{\lim_{x \rightarrow a} (x-a)}_0 + f(a)$$

$$= f(a)$$